

CAPSTONE EXERCISE — Day 3 End-to-End Analytics Workflow			
NovaTech Solutions Sdn. Bhd. — Complete all phases in sequence			
PHASE 1: POWER QUERY — DATA PREPARATION			
No.	Task	Step-by-Step Instructions	Reference
Task 1.1	Load & Clean RAW_Sales	Connect RAW_Sales via Power Query. Steps: (1) Remove 6 duplicate rows (2) Filter out CANCELLED rows (3) Fix date formats — standardise all to YYYY-MM-DD (4) Trim & Proper-case CustomerName (5) Replace 'RM ' prefix in UnitPrice — extract number (6) Replace null Quantity with 0 (7) Remove negative Quantity rows. Load to data model as FACT_Sales_Clean.	RAW_Sales
Task 1.2	Load & Clean RAW_Employees	Connect RAW_Employees. Steps: (1) Proper-case Department column (2) Split FullName into FirstName and LastName columns (3) Replace null Salary_MYR with 4500 (4) Standardise JoinDate to YYYY-MM-DD (5) Add a Tenure_Years calculated column = DateTime.LocalNow() year minus JoinDate year. Load to model as DIM_Employees_Clean.	RAW_Employees
Task 1.3	Unpivot RAW_Expenses	Connect RAW_Expenses. Steps: (1) Select the 12 month columns (Jan-2024 to Dec-2024) (2) Unpivot → produces Attribute and Value columns (3) Rename Attribute to MonthLabel, Value to ActualAmount_MYR (4) Extract MonthNo from MonthLabel (e.g. 'Jan-2024' → 1) (5) Add Year column = 2024. Load as FACT_Expenses_Clean.	RAW_Expenses
Task 1.4	Load Reference Tables	Load RAW_Products as DIM_Products and RAW_Regions as DIM_Regions into the data model without any transformations (they are already clean). Enable 'Load to Data Model'.	RAW_Products, RAW_Regions
Task 1.5	Load DIM_Calendar	Load the DIM_Calendar sheet into the data model. In Power Pivot, go to Design tab → Mark as Date Table → select the 'Date' column. This enables time intelligence DAX functions.	DIM_Calendar
PHASE 2: POWER PIVOT — DATA MODELLING & DAX			
No.	Task	Step-by-Step Instructions	Reference

Task 2.1	Build Star Schema	In Power Pivot Diagram View, create these relationships: • FACT_Sales[Date] → DIM_Calendar[Date] (Active) • FACT_Sales[ProductID] → DIM_Products[ProductID] (Active) • FACT_Sales[SalesRepID] → DIM_Employees[EmployeeID] (Active) • FACT_Expenses[MonthNo] → DIM_Calendar[Month] (Active) Verify all relationships show correct cardinality (Many-to-One).	Power Pivot
Task 2.2	Basic DAX Measures	In the FACT_Sales measure grid, create: [Total Revenue], [Total Qty Sold], [Total Transactions], [Unique Customers], [Avg Order Value], [Total Cost] (using RELATED), [Gross Profit], [Gross Margin %]. Refer to DAX_Guide sheet for exact formulas.	 DAX_Guide
Task 2.3	Time Intelligence Measures	Create in FACT_Sales: [YTD Revenue], [MTD Revenue], [QTD Revenue], [Revenue LY], [YoY Growth %], [Revenue PM], [MoM Growth %]. Test by creating a PivotTable with Month on rows and comparing values.	 DAX_Guide
Task 2.4	CALCULATE Measures	Create segment measures using CALCULATE: [Revenue Enterprise], [Revenue Corporate], [Revenue SME], [Revenue Hardware], [Revenue Software], [% of Total Revenue]. Create [Cumulative Revenue] and [Rolling 3M Revenue].	 DAX_Guide
Task 2.5	Expense DAX Measures	In FACT_Expenses measure grid, create: [Total Actual Expense], [Total Budget], [Budget Variance], [Budget Variance %]. Test using Department slicer.	 DAX_Guide
PHASE 3: REPORTING & DASHBOARD			
No.	Task	Step-by-Step Instructions	Reference
Task 3.1	Revenue by Region Report	New sheet 'Report_Revenue'. PivotTable: Rows = Region Columns = Quarter Values = [Total Revenue], [YoY Growth %]. Add slicer for CustomerSegment. Format revenue as MYR #,##0. Highlight YoY Growth % with data bars.	FACT_Sales

Task 3.2	Product Performance Report	New sheet 'Report_Products'. PivotTable: Rows = Category, SubCategory, ProductName Values = [Total Revenue], [Gross Profit], [Gross Margin %]. Sort Total Revenue Descending. Add Top 10 Value Filter on ProductName. Format Gross Margin % with colour scale.	FACT_Sales
Task 3.3	Monthly Trend Report	New sheet 'Report_Trends'. PivotTable: Rows = Year, Quarter, MonthName (from DIM_Calendar) Values = [Total Revenue], [Revenue LY], [YoY Growth %], [Cumulative Revenue]. Insert Line PivotChart showing Total Revenue vs Revenue LY by month.	DIM_Calendar
Task 3.4	Expense vs Budget Report	New sheet 'Report_Expenses'. PivotTable: Rows = Department, ExpenseCategory Values = [Total Actual Expense], [Total Budget], [Budget Variance], [Budget Variance %]. Apply conditional formatting: Budget Variance > 0 = Red fill, < 0 = Green fill.	FACT_Expenses
Task 3.5	Executive Dashboard	New sheet 'DASHBOARD'. Design: (1) 4 KPI cards — Total Revenue, Gross Margin %, YoY Growth %, Unique Customers (2) Bar chart — Revenue by Region (3) Donut chart — Revenue by Product Category (4) Line chart — Monthly Revenue Trend. Add Region + Segment slicers connected to ALL charts. Apply professional formatting using brand colours.	All Reports
EVALUATION CRITERIA & DELIVERABLES			
No.	Task	Step-by-Step Instructions	Reference
✓ Criterion 1	Clean Data Model	All 5 tables correctly loaded into Power Pivot. Star schema with 4 active relationships. No errors in model.	Phase 1
✓ Criterion 2	DAX Measures	Minimum 15 working DAX measures. Results verified against raw data totals. No BLANK errors.	Phase 2
✓ Criterion 3	PivotTable Reports	4 PivotTable reports — correct fields, proper formatting, working slicers.	Phase 3

✓ Criterion 4	Executive Dashboard	1 polished dashboard with KPI cards, 3 charts, slicers. Professional layout and consistent formatting.	Phase 3
✓ Criterion 5	Data Refresh	Demonstrate: modify 1 row in RAW_Sales → Refresh All → verify dashboard numbers update automatically.	All